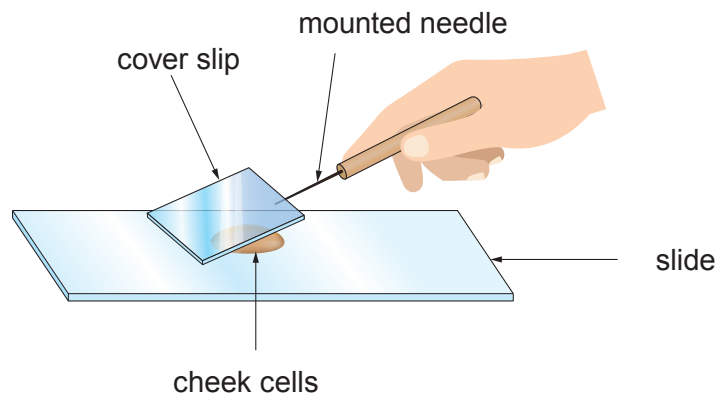


3. (a) Cerys makes a slide of her cheek cells. One stage in her method is shown in **Image 3.1**.

**Image 3.1**



Using the labels on **Image 3.1** describe what Cerys is doing **at this stage of her method**.

[2]

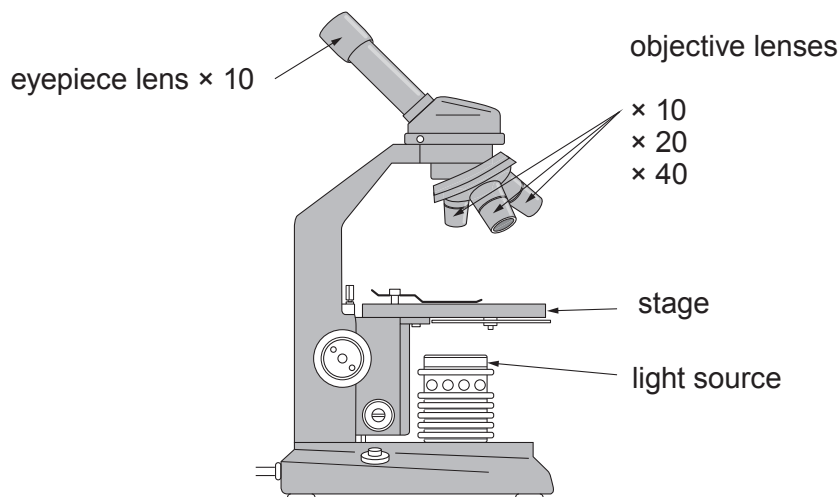
.....

.....

.....

- (b) **Image 3.2** shows a light microscope.

**Image 3.2**



The lowest magnification of the microscope is  $\times 100$ .

Calculate the **highest** magnification of the microscope shown in **Image 3.2**.

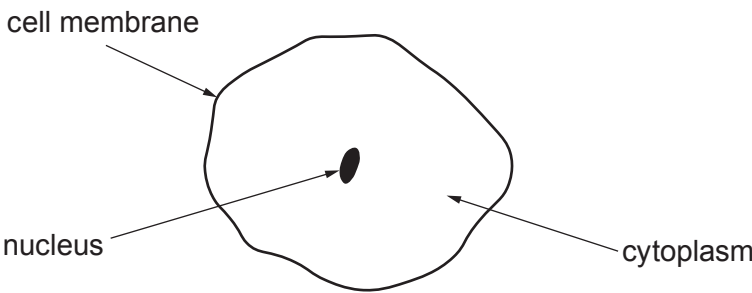
[1]

**Highest magnification** =  $\times$  .....



(c) Cerys drew and labelled a cheek cell. This is shown in **Image 3.3**.

**Image 3.3**



**Complete the table below.** [2]

| Cell structure | Function                    |
|----------------|-----------------------------|
| nucleus        | .....                       |
| .....          | site of most cell reactions |

(d) Cerys put a drop of stain on the slide before adding her cheek cells. [1]  
State the purpose of staining the cells.

.....  
.....

(e) (i) Using this light microscope, it is not possible to see the mitochondria even at the highest magnification. State the reason for this. [1]

.....  
.....

(ii) Complete the following sentence by underlining the correct word from the brackets. [1]

The mitochondria can be seen using an (electronic / electron / electric) microscope.

